

Noah Solomon

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SUMMARY

I am an aspiring hardware and software engineer, with experience ranging from user applications all the way down to the HDL level. I have produced multiple reliable applications; I developed new photon emission microscope control software and used it for FPGA security research, and I overhauled the digital circuits course at Worcester Polytechnic Institute (WPI), introducing a cross-platform Verilator-based interactive simulator.

SKILLS

Languages: Python, C++, Verilog, Assembly, C, SQL, Rust, HTML/CSS, Java, JS, Racket

Software/Tools: PySide6 (Qt), Docker, Vivado (Xilinx), command line, Verilator, Git, MATLAB, Oracle Database

PROJECTS

Interactive FPGA board simulator WPI <https://github.com/TheHarmonicRealm/fpga-sim>

Dec. 2025–

- Redesigned WPI's digital circuits labs, developing new software to simulate users' FPGA code on-screen
- Supported about 120 students' installation and use of my software as a TA
- Programmed a student-friendly terminal interface; refinements include tab completion, validation of user code's I/O ports to provide clear help messages, and heavily filtering Verilator's build errors
- Fully supported all three major operating systems, using Docker and uv for easy installation
- Enabled the creation of unique assignments by letting instructors easily add their own Qt-based boards

Photon emission microscope software WPI Vernam Lab

Mar. 2025–May 2026

- Streamlined the photon emission security research process with a new Qt app for WPI's ALPhANOV microscope
- Designed a more intuitive and responsive GUI to replace the manufacturer's original software
- Implemented reliable Python interfaces for the microscope's subsystems, including camera, stage, and lens switcher
- Developed the software to support another research project, working closely to incorporate that team's feedback; we performed a proof-of-concept extraction attack on an FPGA LLM accelerator, using my software for all imaging
- Won first place in the the ECE department's video competition for my participation in the attack

Les Paul "Sound on Sound" exhibit (3-person team) WPI

Aug.–Oct. 2025

- Designed an exhibit covering Les Paul's overdubbing process for the Waukesha County Historical Society & Museum
- Built a working facsimile of a phonograph and a lathe for realistic live performances of the technique
- Wrote C++ microcontroller code to process audio and drive the device

Historical labeling LLM project (4-person team) WPI, Prague, Czech Republic

Aug.–Dec. 2024

- Tagged 9000 biographies with locations to organize the website of Post Bellum, a historical education nonprofit
- Iteratively developed LLM prompts; used chain-of-thought prompting, with a map applied to clean the final output

EXPERIENCE

Undergraduate tutor, ECE2029 WPI

Mar.–May 2025; Mar.–May 2026

- 2025: Ran weekly lab sessions and graded assignments for the course "Introduction to Digital Circuit Design"
- 2026: Managed the use of my new FPGA board simulation software (see Projects section) and held frequent office hours for tech support, sign-offs, and homework help

LEADERSHIP & ACTIVITIES

Vice President/Social Media Manager, Computer & Technology Club WPI

Feb. 2024–May 2026

- Co-organized weekly educational and social events focused on computer hardware and technology
- Produced social media graphics to advertise club events every week

Captain, Spartronics 4915 (FIRST Robotics Competition) Bainbridge Island, WA Member 2019–2022; Captain 2022

- Led the programming subteam on my school's competitive robotics club, rebuilding after two COVID-affected seasons
- Wrote and tested a substantial portion of our robot's Java code and mentored new programmers
- Contributed to the team's strongest competitive finish in four years, with the team also receiving an award

EDUCATION

Worcester Polytechnic Institute Worcester, MA

Aug. 2022–May 2026

Dual Bachelor of Science in **Computer Science (Cyber Security concentration)** and **Electrical & Computer Engineering**

3.92 GPA • High Distinction • Dean's List 6x • IEEE-HKN honors (ECE) • UPE honors (CS)